

FOUNDATIONS FOR QED II: CLASSICAL THEORY

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ABSTRACT. This paper is the second of a series on the foundations for QED. In the first paper we established the mathematical foundations (see [35]). This paper is devoted to the classical foundations. We first prove the Einstein and Einstein dual compatibility theorems, which shows there always exist two unique inertial frames for the global representation of an interacting system of n particles: one with constant speed of light c and time t relative, and one with relative speed of light b and a unique definition of time τ for all observers. In both cases, we show that a consistent relativistic extension of the special theory to include Newtonian mechanics is possible. We further show that the two approaches are mathematically equivalent but not physically equivalent. We then show that the same holds for classical electrodynamics. We also prove the Wheeler-Feynman conjecture, that classical field theory and delayed action-at-a-distance are different aspects of the same theory, and that the Wheeler-Feynman absorption hypothesis is automatically satisfied without the use of advanced potentials. The dual Maxwell theory contains a longitudinal (dissipative) term in the $\mathbf{E}$ field wave equation, which appears instantaneously with acceleration, and we predict that radiation from a cyclotron will not produce photoelectrons. It is shown that this term provides an effective mass for the photons. A major outcome is the unification of Newtonian mechanics and classical electrodynamics with Einstein's theory, without the need for point particles or self-energy divergence. This means that a second quantized version will not produce a self-energy or infrared divergence. These results, along with the proof of Dyson's first conjecture in [35] resolves all problems with QED.

We use the historical time along with the existence of the 2.7 °K CMBR to show that this frame makes the laws of physics invariant for all observers. We also show that its existence implies that the universe is closed and that quantum mechanics operates on the cosmological scale. We then use historical time to propose a symmetric model for the big bang beginning which corresponds to creation events seen in the laboratory, while avoiding many objections to the current model.

1. Introduction

1.1. The Historical Background and Problems. After the physics community voted in favor of Maxwell's theory and the Lorentz group as the proper transformation theory, one would naturally expect (and many believe) that Newtonian theory is now covered by some version of the Minkowski formulation of the special theory. However, we are now in the

twenty-first century, and the two theories have not been unified. Since Minkowski, many new problems have appeared, and there is no link from a relativistic extension of Newtonian mechanics to QED in sight. There are two major reasons for this. The first is that, if a problem is unsolved for a certain length of time, research in that direction begins to slow down and the problem itself is eventually considered uninteresting. In this atmosphere it becomes a danger to one's professional advancement to investigate these problems. At this point, succeeding generations tend to ignore them as problems or to refer them to a covering theory, which does not solve them.

For example, to by-pass the problem of infinity for a zero-radius particle in the classical electrodynamics (CED) of Lorentz, Dirac replaced the particles with fields. He was then forced to face the infinite field energy at a point (self-energy, see [18] [19]). He then developed an elaborate procedure using both advanced and retarded fields along with a limiting method to obtain a dissipative term, that accounted for radiation reaction as an addition to the Lorentz equation, leading to the Lorentz-Dirac equation. The price paid was an infinite mass renormalization.

This was the motivation for Wheeler and Feynman in their construction of a delayed action-at-a-distance theory, which eliminated the field altogether (see [75]). They were able to solve the divergence problem and obtain the correct radiation reaction term. However, this approach was dropped from favor as it was not quantizable. All the problems with CED (Lorentz or Dirac theory) are still with us but are mostly ignored. At this time, physics education is so decentralized that it's now possible to obtain a Ph.D with concentration in electrodynamics while not knowing the problems exist. Rohrlich [67] is one of the few to even admit that these problems are important and not solved. However, he claims that they belong to the covering theory QED. As one of the early workers on QED, he also knew that QED does not solve them either.

Dirac's version CED was used as the basis for QED. This automatically brought with it the self-energy divergence and added a few others. All these problems were all by-passed in the late 1940s by Feynman, Schwinger, and Tomonaga, leading to the great successes of the era. At the time no one considered this to be the final answer. It was understood that QED was not a stand alone theory. It required the results from the Dirac equation as inputs, with QED providing the last two important pieces. At the time it was believed that the mathematicians would eventually provide the correct mathematical foundations which would make the subtraction of infinities meaningful. After almost fifty years of effort, it became clear that something more than mathematics was required. By that time a new generation had begun to posit string theory as the way forward. However, it has not produced the hoped for outcomes. The electro-weak theory and the standard model each assumed QED as their foundation on the belief the known

problems would be resolved sometime in the near future and, each theory has introduced additional problems.

1.2. Purpose. This paper is the second phase of the first physically consistent and mathematically correct approach to the classical foundations for a direct line of deduction from Newtonian mechanics to quantum electrodynamics. Our purpose in this paper is to obtain the following outcomes:

- (1) A resolution of all problems associated with the unification of Newtonian and Maxwell theories.
- (2) A resolution of the problems associated with radiation reaction, self-energy divergence, field verses particles and advanced interactions in CED.
- (3) Provide classical foundations for: the 2.7 °K cosmic microwave background radiation (CMBR) as a preferred rest frame, for a universal clock, its implications for quantum mechanics and, for a more consistent model of the big bang beginning.

1.3. Summary. In section two, we use the second definition of proper time from the first paper to prove the Einstein and Einstein dual compatibility theorems, which both unify Newton's and Maxwell's theory (see [35]). In section three we extend our results to the general many particle problem and prove that both approaches are mathematical equivalent but not physically equivalent. In section four, we give a more updated view of the photon's mass, speed and wave nature. We also give a refined analysis of relativistic jets seen in our galaxy.

In section five, we use the existence of a universal clock (historical time) along with the 2.7 °K (CMBR) to show that a universal rest frame exists, is available to all observers and the laws of physics are invariant. We show that (CMBR) implies the universe is closed and that quantum mechanics is operative at the cosmic scale. We then couple it with Santilli's isodual numbers to propose a symmetric big bang model, which more closely corresponds to creation events as seen in the laboratory (see [69]).

2. The Einstein-Newtonian Extension

We assume a classical interacting system of n -particles observed by O , who can identify each particle and attach a vector $\mathbf{x}_i$ to the i^{th} particle, denoting its spatial distance to the origin. If $\mathbf{w}_i$ is the velocity of particle i as indicated by O , let $\gamma^{-1}(\mathbf{w}_i) = \sqrt{1 - \mathbf{w}_i^2/c^2}$. The i^{th} particle proper time is defined as

$$d\tau_i = \gamma^{-1}(\mathbf{w}_i)dt, \quad \mathbf{w}_i = \frac{d\mathbf{x}_i}{dt}, \quad d\tau_i^2 = dt^2 - \frac{1}{c^2}d\mathbf{x}_i^2. \quad (2.1)$$

Rewrite the last term to get:

$$dt^2 = d\tau_i^2 + \frac{1}{c^2} d\mathbf{x}_i^2, \Rightarrow cdt = \left(\sqrt{\mathbf{u}_i^2 + c^2} \right) d\tau_i, \quad \mathbf{u}_i = \frac{d\mathbf{x}_i}{d\tau_i} = \gamma(\mathbf{w}_i) \mathbf{w}_i. \quad (2.2)$$

If we let $b_i = \sqrt{\mathbf{u}_i^2 + c^2}$, the second term in Equation (2.2) becomes $cdt = b_i d\tau_i$. This leads to our second definition of proper time:

$$\frac{1}{c} \frac{d}{dt} \equiv \frac{1}{b_i} \frac{d}{d\tau_i}, i = 1, \dots, n. \quad (2.3)$$

This identity provides the correct relationship between the proper time and the observer time for the i^{th} particle. Applying the identity to $\mathbf{x}_i$, we obtain our second identity, which shows that the transformation leaves the configuration (or tangent) space variables invariant for each particle:

$$\frac{\mathbf{w}_i}{c} = \frac{1}{c} \frac{d\mathbf{x}_i}{dt} \equiv \frac{1}{b_i} \frac{d\mathbf{x}_i}{d\tau_i} = \frac{\mathbf{u}_i}{b_i}, i = 1, \dots, n. \quad (2.4)$$

The momentum can be represented as $\mathbf{p}_i = m_i \gamma(\mathbf{w}_i) \mathbf{w}_i = m_i \mathbf{u}_i$, where m_i is the particle rest-mass. Thus, the phase-space variables are also left invariant for each particle. The following theorem, proven in Foundations I, shows that Minkowski's postulate is the cause preventing the relativistic extension of Newtonian mechanics (see [35]).

Theorem 1. (Minkowski Incompatible Theorem) *The addition of Minkowski's postulate to the postulates of Einstein is incompatible for two or more particles.*

This theorem resolved all the confusion caused by the canonical center of mass problem first noted by Pryce [63] and the no-interaction problem first proven by Curry et. al. [16]. The next theorem provides the correct way to make Newtonian mechanics consistent with the special theory.

Theorem 2. (Einstein Compatibility Theorem) *If the observer time is attached to the canonical center of mass and each particle proper time is used, then the global theory is compatibility with the two postulates and the transformations that preserve the first postulate at the particle level are:*

$$b'_i = \gamma(\mathbf{w}) (b_i - \mathbf{u}_i \cdot \mathbf{w}/c), i = 1, \dots, n.$$

and

$$b_i = \gamma(\mathbf{w}) (b'_i + \mathbf{u}'_i \cdot \mathbf{w}/c), i = 1, \dots, n.$$

We can also use the proper time of the canonical center of mass. In this case the, if $\mathbf{V}$ is the velocity of the center of mass as seen by O , it is easy to see that:

$$\frac{1}{c} \frac{d}{dt} \equiv \frac{1}{b} \frac{d}{d\tau} \quad \text{and} \quad \frac{\mathbf{V}}{c} = \frac{\mathbf{U}}{b}, \quad \mathbf{V} = \frac{d\mathbf{X}}{dt}, \quad \mathbf{U} = \frac{d\mathbf{X}}{d\tau}.$$

Where $b = \sqrt{\mathbf{U}^2 + c^2}$, so that $cdt = b d\tau$ and, b is a constant scale factor. This leads to the Einstein dual compatibility theorem:

Theorem 3. (Einstein Dual Compatibility Theorem) *If the proper time of the canonical center of mass is used at the global level and each particle proper time is used locally, then the theory is also compatibility. For the O' observer the global variables are $(\mathbf{X}', \tau)$, $ct' = b'\tau$, where $b' = \sqrt{\mathbf{U}'^2 + c^2}$ and the transformations that preserve the first postulate for the particles are the same:*

$$b'_i = \gamma(\mathbf{v}) (b_i - \mathbf{u}_i \cdot \mathbf{v}/c), i = 1, \dots, n.$$

and

$$b_i = \gamma(\mathbf{v}) (b'_i + \mathbf{u}'_i \cdot \mathbf{v}/c), i = 1, \dots, n,$$

while those that preserve it for the global variables are:

$$b' = \gamma(\mathbf{W}) (b - \mathbf{U} \cdot \mathbf{W}/c) \quad \text{and} \quad b = \gamma(\mathbf{W}) (b' + \mathbf{U} \cdot \mathbf{W}/c).$$

Proof. The first part is clear. We just need to prove that the first postulate is still satisfied. If O' is any other observer who is also investigating the same system of particles with global variables available $(\mathbf{X}', t')$, with relative t' and constant c and $(\mathbf{X}, \tau)$ with relative b' and (the same) τ . Let $\mathbf{W}$ be the relative velocity between O and O' . Because τ is the same for both we only need the relationship between scale factors b and b' to satisfy the first postulate. Because $\mathbf{U} = (\mathbf{X}/\tau)$, we have.

$$\begin{aligned} t' &= \gamma(\mathbf{W}) \left[t - \frac{1}{c^2} (\mathbf{X} \cdot \mathbf{W}) \right] \Rightarrow \frac{b'}{c} \tau = \gamma(\mathbf{W}) \left[\frac{b}{c} \tau - \frac{1}{c^2} (\mathbf{X} \cdot \mathbf{W}) \right] \Rightarrow \\ b' &= \gamma(\mathbf{W}) \left[b - \frac{1}{c} (\frac{\mathbf{X}}{\tau} \cdot \mathbf{W}) \right] = \gamma(\mathbf{W}) \left[b - \frac{1}{c} (\mathbf{U} \cdot \mathbf{W}) \right]. \end{aligned}$$

A similar calculation shows that $b = \gamma(\mathbf{W}) (b' + \mathbf{U}' \cdot \mathbf{W}/c)$. Thus each observer has direct access to all information available to any other observer once they know their relative velocity. Thus Einstein's first postulate is satisfied at the global level. $\square$

Corollary 4. *The two sets of global variables produce mathematically equivalent theories, but not physically equivalent theories.*

2.1. Generator of τ Translations. In the conventional formulation of quantum theory, the Hamiltonian H , is the generator of observer time translations (i.e. t translations). We choose a direct canonical approach to our derivation of the generator for τ translations for a number of reasons:

- (1) It is simple, and, directly related to the non-relativistic case.
- (2) It maintains mathematical equivalence of the classical equations of motion.
- (3) It provides a direct road from the classical theory in Poisson brackets to the quantum theory in Heisenberg brackets (i.e., $\{ \cdot \} \rightarrow \frac{1}{i\hbar} [\cdot]$)
- (4) The quantized version is (will be) strictly positive definite so that a particle interpretation of the equations is possible.
- (5) Because the mass is fixed, along with the spin, at the quantum level it is the natural choice to label the irreducible representations of the (proper-time) Poincaré algebra describing elementary particles.

To see how this may be done, let W be any classical phase space observable so that the Poisson bracket defines Hamilton's equations in the O frame by:

$$\frac{dW}{dt} = \{H, W\} = \frac{\partial H}{\partial \mathbf{p}} \frac{\partial W}{\partial \mathbf{x}} - \frac{\partial H}{\partial \mathbf{x}} \frac{\partial W}{\partial \mathbf{p}}.$$

We also have $cdt = b d\tau$, so that $\frac{dt}{d\tau} = \frac{b}{c}$. To represent the dynamics via the proper time of the particle of rest mass m , the Hamiltonian can be represented as $H = mc^2 \gamma(\mathbf{w})$, so that $\gamma^{-1}(\mathbf{w}) = mc^2/H$. Thus, we can also represent the proper time by $d\tau = (mc^2/H)dt$. Using the chain rule, we have:

$$\frac{dW}{d\tau} = \frac{dt}{d\tau} \frac{dW}{dt} = \frac{b}{c} \frac{dW}{dt} = \frac{H}{mc^2} \{H, W\}. \quad (2.5)$$

Using the invariant rest energy mc^2 , we determine the canonical proper-time Hamiltonian K such that

$$\{K, W\} = \frac{H}{mc^2} \{H, W\}, \quad K|_{\mathbf{p}=0} = H|_{\mathbf{p}=0} = mc^2.$$

From

$$\begin{aligned} \{K, W\} &= \left[\frac{H}{mc^2} \frac{\partial H}{\partial \mathbf{p}} \right] \frac{\partial W}{\partial \mathbf{x}} - \left[\frac{H}{mc^2} \frac{\partial H}{\partial \mathbf{x}} \right] \frac{\partial W}{\partial \mathbf{p}} \\ &= \frac{\partial}{\partial \mathbf{p}} \left[\frac{H^2}{2mc^2} + a \right] \frac{\partial W}{\partial \mathbf{x}} - \frac{\partial}{\partial \mathbf{x}} \left[\frac{H^2}{2mc^2} + a' \right] \frac{\partial W}{\partial \mathbf{p}}, \end{aligned}$$

we see that $a = a' = \frac{1}{2}mc^2$. Thus, assuming no explicit time dependence, we have:

$$K = \frac{H^2}{2mc^2} + \frac{mc^2}{2} = \frac{\mathbf{p}^2}{2m} + mc^2, \quad \text{and} \quad \frac{dW}{d\tau} = \{K, W\}. \quad (2.6)$$

Remark 5. We included the third term in equation (2.5) to show that the dual (or mathematically equivalent) relationship is manifest from the beginning.

Equation (2.6) was first derived by Gill and Lindesay [40]. Although it looks like the non-relativistic case, it is fully relativistic and (partially) eliminates the problems associated with the square root in the conventional implementation. The most general solution is as follows.

$$K = mc^2 + \int_{mc^2}^H (dt/d\tau) d\bar{H} = mc^2 + \int_{mc^2}^H (\bar{H}/mc^2) d\bar{H} \quad (2.7)$$

There are three possible solutions to this equation depending on the assumptions made.

(1) By keeping the Lorentz frame fixed with H/mc^2 constant we obtain

$$K = \frac{H^2}{mc^2} = \frac{\mathbf{p}^2}{mc^2} + mc^2. \quad (2.8)$$

This form was first derived by Gill [32], and used to provide a particle representation for the Klein-Gordon equation with positive probability density and with the proper-time as an operator.

- (2) Keeping the mass fixed and allowing the Lorentz frame to vary (boost), we obtain Equation (2.6).
- (3) Keeping the momentum P fixed and allowing the Lorentz frame, H and the mass M to vary, we obtain

$$K = Mc^2 = \sqrt{H^2 - c^2 \mathbf{P}^2}. \quad (2.9)$$

This is the appropriate Hamiltonian in the constant-momentum frame. This form has received the most attention and has been used to associate the source proper time with the (off-shell) mass operator in parametrized relativistic quantum theories (see Fanchi in [27]).

2.2. Einstein-Newtonian Particle.

2.2.1. First Case. For the first relativistic extension of Newton's theory we take the one that eliminates the square root:

$$H = \sqrt{c^2 \mathbf{p}^2 + (mc^2 + V)^2}.$$

Direct calculation provides:

$$K = \frac{\mathbf{p}^2}{2m} + V + \frac{V^2}{2mc^2} + mc^2. \quad (2.10)$$

This equation differs from $H = \frac{\mathbf{p}^2}{2m} + V$ by the last two terms. In order to see what new physics these terms introduce, let $V = -mc^2 r_0/r$. From Hamilton's equations, we obtain:

$$\mathbf{u} = \frac{\partial K}{\partial \mathbf{p}} = \frac{\mathbf{p}}{m} \quad \text{and} \quad \mathbf{F}_K = -\frac{\partial K}{\partial \mathbf{x}} = -\nabla V \left(1 + \frac{V}{mc^2} \right). \quad (2.11)$$

Because $V/mc^2 = -r_0/r$, we see that:

- (1) $\mathbf{F}_K$ is attractive when $r > r_0$,
- (2) $\mathbf{F}_K = 0$ when $r = r_0$ and
- (3) $\mathbf{F}_K$ is repulsive when $r < r_0$.

If we compute Hamilton's equations using H , we obtain:

$$\begin{aligned} \mathbf{w} &= \frac{\partial H}{\partial \mathbf{p}} = \frac{c^2 \mathbf{p}}{H} \quad \text{and} \\ \mathbf{F}_H &= -\frac{\partial H}{\partial \mathbf{x}} = -\nabla V \frac{(mc^2 + V)}{H} = \frac{mc^2}{H} \left[-\nabla V \left(1 + \frac{V}{mc^2} \right) \right] = \frac{c}{b} \mathbf{F}_K. \end{aligned}$$

2.2.2. Second Case. For the second case we take $H = \sqrt{c^2 \mathbf{p}^2 + m^2 c^4} + V$. In this case, K becomes:

$$K = \frac{H^2}{2mc^2} + \frac{mc^2}{2} = \frac{\mathbf{p}^2}{2m} + mc^2 + \frac{V^2}{2mc^2} + \frac{V\sqrt{c^2 \mathbf{p}^2 + m^2 c^4}}{mc^2}.$$

If we set $H_0 = \sqrt{c^2 \mathbf{p}^2 + m^2 c^4}$ and compute Hamilton's equations, we obtain:

$$\frac{d\mathbf{x}}{d\tau} = \frac{\partial K}{\partial \mathbf{p}} = \frac{H}{mc^2} \left(\frac{c^2 \mathbf{p}}{H_0} \right) = \frac{b}{c} \left(\frac{c^2 \mathbf{p}}{H_0} \right) = \frac{b}{c} \frac{d\mathbf{x}}{dt} \quad (2.12)$$

and

$$\frac{d\mathbf{p}}{d\tau} = -\frac{b}{c} \nabla V \left[1 + \frac{V}{H_0} \right]. \quad (2.13)$$

In both cases the standard and dual equations of motion are mathematically equivalent and have the related but not the same physical properties.

3. EINSTEIN AND EINSTEIN DUAL THEORY

Returning to the many-particle clock problem, recall we assumed n interacting particles have Hamiltonians H_i and total Hamiltonian $H = \sum_{i=1}^n H_i$. The effective mass energy Mc^2 , and total momentum $\mathbf{P}$, was defined as follows:

$$Mc^2 = \sqrt{H^2 - c^2 \mathbf{P}^2}, \quad \mathbf{P} = \sum_{i=1}^n \mathbf{p}_i.$$

We can now represent the Hamiltonian by $H = \sqrt{c^2 \mathbf{P}^2 + M^2 c^4}$.

As noted previously, for the O frame observer, the canonical center of mass position $\mathbf{X}$ is defined by (see [49])

$$\mathbf{X} = \frac{1}{H} \sum_{i=1}^n H_i \mathbf{x}_i + \frac{c^2 (\mathbf{S} \times \mathbf{P})}{H (Mc^2 + H)}, \quad (3.1)$$

where $\mathbf{S}$ is the global spin of the system of particles relative to O . (It is clear that (3.1) cannot represent the vector part of a four-vector.) If there is no interaction, S, H and M are constant, and independent of the $\{\mathbf{x}_i, \mathbf{p}_i\}$ variables, so that:

$$\{X_i, X_j\} = \sum_{k=1}^n \frac{\partial X_i}{\partial \mathbf{p}_k} \cdot \frac{\partial X_j}{\partial \mathbf{x}_k} - \frac{\partial X_j}{\partial \mathbf{p}_k} \cdot \frac{\partial X_i}{\partial \mathbf{x}_k} \equiv 0.$$

However, when interaction is present, S, H and M may all depend on the $\{\mathbf{x}_i, \mathbf{p}_i\}$ variables, so that in general $\{X_i, X_j\} \neq 0$. Because $\mathbf{X}$ is the canonical conjugate of $\mathbf{P}$, it is precisely what we need for a consistent merge with quantum mechanics.

Let $\mathbf{V}$ be the velocity of $\mathbf{X}$ with respect to O . It follows that H also has the representation $H = Mc^2 \gamma(\mathbf{V})$, such that $\gamma(\mathbf{V})^{-1} = (Mc^2/H)$. In this representation, we observe that $d\tau = \gamma(\mathbf{V})^{-1} dt = (Mc^2/H) dt$ does not depend on the number of particles in the system. It follows that, as long as

Mc^2/H is fixed, τ is invariant, so that the number of particles can increase or decrease without changing τ . (This means that number n is not conserved and, in some cases of physical interest, may even be a integer-valued random variable).

From $dt^2 = d\tau^2 + d\mathbf{X}^2/c^2$, we see that ($\mathbf{U} = d\mathbf{X}/d\tau$)

$$c^2 dt^2 = (c^2 + \mathbf{U}^2) d\tau^2 \Rightarrow c dt = \left(\sqrt{c^2 + \mathbf{U}^2} \right) d\tau.$$

It is easy to see that $\mathbf{U} = \gamma(\mathbf{V})\mathbf{V}$, such that $\mathbf{U}$ is constant. If we define $b = \sqrt{\mathbf{U}^2 + c^2}$, then we can write $c dt = b d\tau$. Because b is constant $ct = b\tau$. For observer O' the same system has velocity $\mathbf{V}'$ for the center of mass and, by the same calculations, we obtain $ct' = b'\tau$, where $b' = \sqrt{\mathbf{U}'^2 + c^2}$. This shows that a unique (operational) measure of time is available to all the observers. Furthermore, τ differs from t (t') by a constant scale factor. Thus, all observers may uniquely define the local-time of the center of mass for the system of particles (independent of their chosen inertial frame). We also obtain the following identity:

$$\frac{1}{c} \frac{d}{dt} \equiv \frac{1}{b} \frac{d}{d\tau} \equiv \frac{1}{b_i} \frac{d}{d\tau_i}, i = 1, \dots, n. \quad (3.2)$$

Applying the above to $\mathbf{x}_i$ we see that:

$$\frac{1}{c} \frac{d\mathbf{x}_i}{dt} \equiv \frac{1}{b} \frac{d\mathbf{x}_i}{d\tau} \equiv \frac{1}{b_i} \frac{d\mathbf{x}_i}{d\tau_i}, i = 1, \dots, n. \quad (3.3)$$

Theorem 6. *Two sets of global variables are available to O : $(\mathbf{X}, t)$ and $(\mathbf{X}, \tau)$. The use of $(\mathbf{X}, t)$ provides a relative definition of time and constant speed of light, while the use of $(\mathbf{X}, \tau)$ provides a unique definition of time and relative definition of the speed of light.*

Proof. The first part is clear. We just need to prove that the first postulate is still satisfied. If O' is any other observer who is also investigating the same system of particles with global variables available $(\mathbf{X}', t')$, with relative t' and constant c and $(\mathbf{X}', \tau)$ with relative b' and constant τ . Let $\mathbf{W}$ be the relative velocity between O and O' . Because τ is the same for both we only need the relationship between scale factors b and b' to satisfy the first postulate. Because $\mathbf{U} = (\mathbf{X}/\tau)$, we have.

$$\begin{aligned} t' &= \gamma(\mathbf{W}) \left[t - \frac{1}{c^2} (\mathbf{X} \cdot \mathbf{W}) \right] \Rightarrow \frac{b'}{c} \tau = \gamma(\mathbf{W}) \left[\frac{b}{c} \tau - \frac{1}{c^2} (\mathbf{X} \cdot \mathbf{W}) \right] \Rightarrow \\ b' &= \gamma(\mathbf{W}) \left[b - \frac{1}{c} \left(\frac{\mathbf{X}}{\tau} \cdot \mathbf{W} \right) \right] = \gamma(\mathbf{W}) \left[b - \frac{1}{c} (\mathbf{U} \cdot \mathbf{W}) \right]. \end{aligned}$$

A similar calculation shows that $b = \gamma(\mathbf{W}) (b' + \mathbf{U}' \cdot \mathbf{W}/c)$. Thus each observer has direct access to all information available to any other observer once they know their relative velocity. Thus Einstein's first postulate is satisfied at the global level. $\square$

Corollary 7. *The two sets of global variables produce mathematically equivalent theories, but not physically equivalent theories.*

Theorem 8. *If we use the proper time for each particle, then the special theory of Einstein holds for any many-particle system.*

Remark 9. *It is important to note that equation (3.3) creates serious problems for any assumption, except by direct measurement, that the speed of light is c and or the velocity of distant objects are traveling with speeds less than c . For example, in all astronomical cases, we can only measure the ratio β and assume that it equals $\frac{v}{c}$. However, because it is possible that $\beta = \frac{u}{b}$, the object could also have speed $u > c$ and we are looking at light traveling with speed $b \gg c$. It follows that serious retrospection is required in both astronomy and cosmology.*

3.1. Global Theory. If we treat the system of particles as a single entity, then $(\mathbf{X}, \mathbf{P})$ are the natural phase space variables for the global external system dynamics.

3.2. $(\mathbf{X}, t)$. In this case, $H = \sqrt{c^2 \mathbf{P}^2 + M^2 c^4}$ and from Hamilton's equation for the velocity, we obtain in the O frame:

$$\mathbf{V} = \frac{\partial H}{\partial \mathbf{P}} = \frac{c^2 \mathbf{P}}{H} = \frac{c^2}{H} \sum_{i=1}^n \mathbf{p}_i = \frac{c^2}{H} \sum_{i=1}^n m_i \frac{d\mathbf{x}_i}{d\tau_i},$$

while in the O' frame we obtain:

$$\mathbf{V}' = \frac{\partial H'}{\partial \mathbf{P}'} = \frac{c^2 \mathbf{P}'}{H'} = \frac{c^2}{H'} \sum_{i=1}^n \mathbf{p}'_i = \frac{c^2}{H'} \sum_{i=1}^n m_i \frac{d\mathbf{x}'_i}{d\tau'_i}.$$

We focus on the O frame but the same calculation applies to the O' frame. Using $\frac{H_i}{m_i c^2} d\tau_i = dt$, we have:

$$\mathbf{X} = \frac{1}{H} \sum_{i=1}^n H_i \mathbf{x}_i + \mathbf{Y} \quad \text{and} \quad \mathbf{X}' = \frac{1}{H'} \sum_{i=1}^n H'_i \mathbf{x}'_i + \mathbf{Y}'.$$

Where $\mathbf{Y}$ and $\mathbf{Y}'$ are constants of integration. Thus, we obtain the canonical center of mass up to a constant.

3.3. $(\mathbf{X}, \tau)$. For the dual case, the time evolution of W with respect to τ is defined by:

$$K = \frac{H^2}{2Mc^2} + \frac{Mc^2}{2} = \frac{\mathbf{P}^2}{2M} + Mc^2, \quad \text{and} \quad \frac{dW}{d\tau} = \{K, W\}. \quad (3.4)$$

As K does not depend on the center-of-mass position $\mathbf{X}$, it is easy to see that

$$\mathbf{U} = \frac{d\mathbf{X}}{d\tau} = \frac{\partial K}{\partial \mathbf{P}} = \frac{\mathbf{P}}{M} = \frac{1}{M} \sum_{i=1}^n \mathbf{p}_i = \frac{1}{M} \sum_{i=1}^n m_i \mathbf{u}_i = \frac{1}{M} \sum_{i=1}^n m_i \frac{d\mathbf{x}_i}{d\tau_i}. \quad (3.5)$$

For the O' observer, the same calculation leads to:

$$\mathbf{U}' = \frac{d\mathbf{X}'}{d\tau} = \frac{\partial K}{\partial \mathbf{P}'} = \frac{\mathbf{P}'}{M'} = \frac{1}{M'} \sum_{i=1}^n \mathbf{p}'_i = \frac{1}{M'} \sum_{i=1}^n m'_i \mathbf{u}'_i = \frac{1}{M'} \sum_{i=1}^n m'_i \frac{d\mathbf{x}'_i}{d\tau'_i}. \quad (3.6)$$

We now observe that

$$dt = \frac{H_i}{m_i c^2} d\tau_i = \frac{H}{M c^2} d\tau \quad \Rightarrow \quad \frac{m_i}{M} \frac{d}{d\tau_i} = \frac{H_i}{H} \frac{d}{d\tau}.$$

Thus, we can replace (3.5) and (3.6) by:

$$\frac{d\mathbf{X}}{d\tau} = \frac{\partial K}{\partial \mathbf{P}} = \frac{\mathbf{P}}{M} = \frac{1}{M} \sum_{i=1}^n \mathbf{p}_i = \frac{1}{M} \sum_{i=1}^n m_i \mathbf{u}_i = \frac{1}{H} \sum_{i=1}^n H_i \frac{d\mathbf{x}_i}{d\tau} \quad (3.7)$$

and

$$\frac{d\mathbf{X}'}{d\tau} = \frac{\partial K}{\partial \mathbf{P}'} = \frac{\mathbf{P}'}{M'} = \frac{1}{M'} \sum_{i=1}^n \mathbf{p}'_i = \frac{1}{M'} \sum_{i=1}^n m'_i \mathbf{u}'_i = \frac{1}{H'} \sum_{i=1}^n H'_i \frac{d\mathbf{x}'_i}{d\tau}. \quad (3.8)$$

Because the H_i (respectively H'_i) does not depend on τ , we can integrate both equations to get:

$$\mathbf{X} = \frac{1}{H} \sum_{i=1}^n H_i \mathbf{x}_i + \mathbf{Y} \quad \text{and} \quad \mathbf{X}' = \frac{1}{H'} \sum_{i=1}^n H'_i \mathbf{x}'_i + \mathbf{Y}',$$

where once again, $\mathbf{Y}$ and $\mathbf{Y}'$ are the constants of integration. Thus, in both cases, Hamiltonian determines the canonical position up to a constant value.

To see directly that the clock transformation is also a canonical change of variables (time), which leaves the phase space invariant, we have the following theorem:

Theorem 10. *There exists a function $S = S(\mathbf{X}, \mathbf{P}, \tau)$ such that*

$$\mathbf{P} \cdot d\mathbf{X} - H dt \equiv \mathbf{P} \cdot d\mathbf{X} - K d\tau + dS.$$

Proof. Set $S = [K - M c^2] \tau$. An easy calculation, using the fact that both $M c^2$ and K are conserved quantities, shows that $dS = [K - M c^2] d\tau$. Because τ is an invariant, $d\tau$ is an exact form, so that dS is a total differential. $\square$

We also have that

$$\sum_{i=1}^n [\mathbf{p}_i \cdot d\mathbf{x}_i - H_i dt] = \sum_{i=1}^n \mathbf{p}_i \cdot d\mathbf{x}_i - \sum_{i=1}^n H_i dt = \sum_{i=1}^n \mathbf{p}_i \cdot d\mathbf{x}_i - H dt.$$

This result leads to:

Corollary 11. *There exists a function $S = S(\{\mathbf{x}_i\}, \{\mathbf{p}_i\}, \tau)$ such that*

$$\sum_{i=1}^n \mathbf{p}_i \cdot d\mathbf{x}_i - H dt \equiv \sum_{i=1}^n \mathbf{p}_i \cdot d\mathbf{x}_i - K d\tau + dS. \quad (3.9)$$

3.4. The Clock Relationships. The global observer has two possible clocks to use, so we need to understand the basic relationships between the global system clock and the clocks of the individual particles for the two cases.

3.4.1. The $(\mathbf{X}, t)$ Relationship. Let $W(\mathbf{X}, t)$ be any observable. In this case $\frac{H_i}{m_i c^2} d\tau_i = dt$ and

$$H = \sqrt{c^2 P^2 + M^2 c^4} = \sum_{i=1}^n H_i \quad \text{with} \quad H_i = \sqrt{c^2 \pi_i^2 + m_i^2 c^4} + V_i.$$

Thus,

$$\begin{aligned} \frac{dW}{dt} &= \{H, W\} = \sum_{i=1}^n \{H_i, W\} = \sum_{i=1}^n \frac{m_i c^2}{H_i} \frac{H_i}{m_i c^2} \{H_i, W\} \\ &= \sum_{i=1}^n \frac{m_i c^2}{H_i} \{K_i, W\} = \sum_{i=1}^n \frac{d\tau_i}{dt} \{K_i, W\}, \end{aligned} \quad (3.10)$$

3.4.2. The $(\mathbf{X}, \tau)$ Relationship. In this case, we know that $d\tau_i/d\tau = H m_i / H_i M$ and for $W(\mathbf{X}, \tau)$, we obtain:

$$\begin{aligned} \frac{dW}{d\tau} &= \{K, W\} = \frac{H}{M c^2} \{H, W\} = \frac{H}{M c^2} \sum_{i=1}^n \{H_i, W\} \\ &= \frac{H}{M c^2} \sum_{i=1}^n \frac{m_i c^2}{H_i} \left[\frac{H_i}{m_i c^2} \{H_i, W\} \right] = \sum_{i=1}^n \frac{H m_i}{H_i M} \{K_i, W\} = \sum_{i=1}^n \frac{d\tau_i}{d\tau} \{K_i, W\}. \end{aligned} \quad (3.11)$$

From both equations (3.10) and (3.11), we see that:

$$dW = \sum_{i=1}^n d\tau_i \{K_i, W\}. \quad (3.12)$$

This is to be expected because t and τ differ by a constant scale factor (i.e., $cdt = b d\tau$). This may seem trivial but it is not, because $b \geq c$, with equality if and only if the observer can locate his frame at the center of mass of the system of particles.

3.4.3. Clock Relationship and Lagrangian. For the Lagrangian, we return to our definition of the global Hamiltonian K and observable W . In this case, we know that

$$d\tau \{K, W\} \equiv \sum_{i=1}^n d\tau_i \{K_i, W\}. \quad (3.13)$$

Using equation (3.9), we obtain:

$$\begin{aligned} \mathcal{L} dt &= \sum_{i=1}^n \mathbf{p}_i \cdot d\mathbf{x}_i - H dt = \sum_{i=1}^n \mathbf{p}_i \cdot d\mathbf{x}_i - K d\tau + dS \\ &= \sum_{i=1}^n [\mathbf{p}_i \cdot d\mathbf{x}_i - K_i d\tau_i] + dS = \sum_{i=1}^n \mathcal{L}_i d\tau_i + dS. \end{aligned} \quad (3.14)$$

Theorem 12. *Any closed system of interacting particles has two distinct sets of inertial frame coordinates for each observer to describe each particle and the system as a whole. The following holds:*

- (1) In one frame, the speed of light c is an invariant upper bound and time t is relative, whereas in the other, the time τ is invariant and the speed of light b , is relative.
- (2) For the entire system and each particle, the equations of motion are mathematically equivalent.

We have already proven all but the last part of the above theorem, which will be completed when we study the Maxwell theory.

Definition 13. A theory with two mathematically equivalent representations is said to be Einsteinian if at least one satisfies the two postulates of the special theory.

3.5. Global Interaction: At the global level, it suffices to assume that the interaction is via a potential $V(\mathbf{X})$. We can add V to the equation for H , to obtain:

$$\begin{aligned} H &= \sqrt{c^2 \mathbf{P}^2 + M^2 c^4} + V(\mathbf{X}) = H_0 + V(\mathbf{X}) \Rightarrow \\ \frac{d\mathbf{X}}{dt} &= \frac{c^2 \mathbf{P}}{H_0} \quad \text{and} \quad \frac{d\mathbf{P}}{dt} = -\nabla V(\mathbf{X}). \end{aligned} \quad (3.15)$$

For comparison, if we use the proper clock, we get:

$$\begin{aligned} K &= \frac{H^2}{2Mc^2} + \frac{Mc^2}{2} \Rightarrow \frac{d\mathbf{X}}{d\tau} = \frac{\partial K}{\partial \mathbf{P}} = \frac{H}{Mc^2} \frac{c^2 \mathbf{P}}{H_0} = \frac{b}{c} \frac{d\mathbf{X}}{dt}, \\ \frac{d\mathbf{P}}{d\tau} &= \frac{\partial K}{\partial \mathbf{X}} = -\frac{H}{Mc^2} \nabla V(\mathbf{X}) = \frac{b}{c} \frac{d\mathbf{P}}{dt}. \end{aligned} \quad (3.16)$$

A comparison of (3.15) and (3.16) shows that the two clocks provide mathematically equivalent equations of motion for the general system dynamics, which are not physically equivalent.

3.6. Dual Maxwell Theory.

3.6.1. Dynamics of a Particle. We now investigate the corresponding single particle dynamical theory. In this section, $b = b_i$, $\tau = \tau_i$ and $\mathbf{u} = \mathbf{u}_i$.

For the relativistic extension of Maxwell's theory we use $\sqrt{c^2 \mathbf{p}^2 + m^2 c^4} \rightarrow \sqrt{c^2 \boldsymbol{\pi}^2 + m^2 c^4} + V$, where $\mathbf{A}$ is the vector potential, $V = e\Phi$ is the potential energy, $\mathbf{E} = -\frac{1}{b}(\partial \mathbf{A} / \partial \tau) - \nabla \Phi$ and $\boldsymbol{\pi} = \mathbf{p} - \frac{e}{c} \mathbf{A}$. In this case, K becomes:

$$K = \frac{H^2}{2mc^2} + \frac{mc^2}{2} = \frac{\boldsymbol{\pi}^2}{2m} + mc^2 + \frac{V^2}{2mc^2} + \frac{V\sqrt{c^2 \boldsymbol{\pi}^2 + m^2 c^4}}{mc^2}.$$

If we set $H_0 = \sqrt{c^2 \boldsymbol{\pi}^2 + m^2 c^4}$, use standard vector identities with $\nabla \times \boldsymbol{\pi} = -\frac{e}{c} \mathbf{B}$, and compute Hamilton's equations, we obtain:

$$\frac{d\mathbf{x}}{d\tau} = \frac{\partial K}{\partial \mathbf{p}} = \frac{H}{mc^2} \left(\frac{c^2 \boldsymbol{\pi}}{H_0} \right) = \frac{b}{c} \left(\frac{c^2 \boldsymbol{\pi}}{H_0} \right) = \frac{b}{c} \frac{d\mathbf{x}}{dt} \quad (3.17)$$

and

$$\begin{aligned}
\frac{d\mathbf{p}}{d\tau} &= \frac{b}{c} \frac{[(c^2\boldsymbol{\pi} \cdot \boldsymbol{\nabla}) \mathbf{A} + \frac{e}{b} (c^2\boldsymbol{\pi} \times \mathbf{B})]}{H_0} - \frac{b}{c} \boldsymbol{\nabla} V \\
&= \frac{b}{c} [(\mathbf{u} \cdot \boldsymbol{\nabla}) \mathbf{A} + \frac{e}{b} (\mathbf{u} \times \mathbf{B})] - \frac{b}{c} \boldsymbol{\nabla} V \\
&= \frac{b}{c} \left[e\mathbf{E} + \frac{e}{b} (\mathbf{u} \times \mathbf{B}) + \frac{e}{b} \frac{d\mathbf{A}}{d\tau} \right] \Rightarrow \\
\frac{c}{b} \frac{d\boldsymbol{\pi}}{d\tau} &= [e\mathbf{E} + \frac{e}{b} (\mathbf{u} \times \mathbf{B})] = \frac{d\boldsymbol{\pi}}{dt}.
\end{aligned} \tag{3.18}$$

Equations (3.17) and (3.18) indicate that the standard and dual equations of motion are mathematically equivalent. Thus, the assumption that K remains invariant with minimal coupling is valid. This also completes the proof of Theorem 12 (mathematical equivalence).

3.6.2. Non-Physical Equivalence. To see that the theory is not physically equivalent, set $H_0 = \sqrt{c^2\boldsymbol{\pi}^2 + m^2c^4}$, use standard vector identities with $H_0 = mcb$, $\boldsymbol{\nabla} \times \boldsymbol{\pi} = -\frac{e}{c}\mathbf{B}$, and compute Hamilton's equations (again) in more detail to obtain:

$$\begin{aligned}
\mathbf{u} &= \frac{d\mathbf{x}}{d\tau} = \left[1 + \frac{V}{H_0} \right] \frac{\boldsymbol{\pi}}{m} = \frac{\boldsymbol{\pi}}{\tilde{m}}, \quad \tilde{m} = \left[1 + \frac{V}{H_0} \right]^{-1} m, \\
\frac{d\mathbf{p}}{d\tau} &= - \frac{[(\boldsymbol{\pi} \cdot \boldsymbol{\nabla}) \boldsymbol{\pi} + \frac{e}{c} \boldsymbol{\pi} \times \mathbf{B}]}{m} \left[1 + \frac{V}{H_0} \right] - \boldsymbol{\nabla} V \frac{H_0}{mc^2} \left[1 + \frac{V}{H_0} \right] \\
&= \frac{e}{c} (\mathbf{u} \cdot \boldsymbol{\nabla}) \mathbf{A} + \frac{e}{c} \mathbf{u} \times \mathbf{B} - \boldsymbol{\nabla} V \frac{b}{c} \left[1 + \frac{V}{H_0} \right].
\end{aligned} \tag{3.19}$$

We remark that one can view $\tilde{m}$ as a (finite) renormalization of m which occurs the moment that the potential is turned on. This may seem strange to one not familiar with the history of this subject. An excellent discussion of renormalization from a historical perspective can be found in the article by Dresden [20].

If we multiply equation (3.19) by b/c , compute the total derivative of $\mathbf{A}$ with respect to τ and use the definition of $\mathbf{E}$, with $V = e\Phi$, we have:

$$\begin{aligned}
\left[\frac{d\mathbf{p}}{d\tau} - \frac{e}{c} \frac{d\mathbf{A}}{d\tau} \right] &= -\frac{e}{c} \frac{\partial \mathbf{A}}{\partial \tau} + \frac{e}{c} \mathbf{u} \times \mathbf{B} - \frac{be}{c} \boldsymbol{\nabla} \Phi \left[1 + \frac{V}{mcb} \right] \\
&= e\mathbf{E} + \frac{e}{c} \mathbf{u} \times \mathbf{B} - e\boldsymbol{\nabla} \Phi \frac{V}{mc^2}.
\end{aligned} \tag{3.20}$$

The last term in equation (3.20) is an addition to the Lorentz force, with the opposite sign of $-\boldsymbol{\nabla} \Phi$, which appears in $\mathbf{E}$. In order to see the physical meaning of the term, assume an interaction between a proton and an electron, where $\mathbf{A} = \mathbf{0}$ and V is the Coulomb interaction, so that (3.19) and

(3.20) become:

$$\mathbf{u} = \left[1 + \frac{V}{mc^2} \right] \frac{\pi}{m},$$

$$\frac{c}{b} \frac{d\mathbf{p}}{d\tau} = -\nabla V - \nabla V \frac{V}{mcb}.$$

If we treat $\mathbf{u}$ as (approximately) $\mathbf{p}/m$ and set $b = c$, we get that

$$m\mathbf{a} = -\nabla V - \nabla V \frac{V}{mc^2}.$$

In this case, it is easy to show that the classical electron radius, r_0 , is a critical point (i.e., $-\nabla\Phi - \nabla\Phi(V/mc^2) = 0$). Thus, for $0 < r < r_0$, the force becomes repulsive. We interpret this as a fixed region of repulsion, so that the singularity $r = 0$ is impossible to reach at the classical level. The neglected terms are attractive but of lower order. This makes the critical point slightly less than r_0 . Thus, in general, the electron experiences a strongly repulsive force when it gets too close to the proton. This means that the classical principle of impenetrability, namely that no two particles can occupy the same space at the same time occurs naturally. Furthermore, this analysis shows conclusively that information about the (classical) structure of the particle is not required in the dual theory. This suggests that an electron may reach an unstable bound state inside the proton. (Recall the Maxwell's equations are already spin-1 relativistic wave equations.)

Remark 14. Referring back to section 2.2, we see that the above is not a special case and applies to any attractive potential.

3.6.3. Field of a Particle. To study the field of a particle, we write Maxwell's equations (in c.g.s. units):

$$\begin{aligned} \nabla \cdot \mathbf{B} &= 0, & \nabla \cdot \mathbf{E} &= 4\pi\rho, \\ \nabla \times \mathbf{E} &= -\frac{1}{c} \frac{\partial \mathbf{B}}{\partial t}, & \nabla \times \mathbf{B} &= \frac{1}{c} \left[\frac{\partial \mathbf{E}}{\partial t} + 4\pi\rho\mathbf{w} \right]. \end{aligned} \quad (3.21)$$

Using equations (2.3) and (2.4) in (3.21), we obtain a (*mathematically identical representation*):

$$\begin{aligned} \nabla \cdot \mathbf{B} &= 0, & \nabla \cdot \mathbf{E} &= 4\pi\rho, \\ \nabla \times \mathbf{E} &= -\frac{1}{b} \frac{\partial \mathbf{B}}{\partial \tau}, & \nabla \times \mathbf{B} &= \frac{1}{b} \left[\frac{\partial \mathbf{E}}{\partial \tau} + 4\pi\rho\mathbf{u} \right]. \end{aligned} \quad (3.22)$$

Thus, we obtained a mathematically equivalent set of Maxwell's equations using the local time of the particle to describe its field.

To derive the corresponding wave equations, we take the curl of the last two equations in (3.22), and use $\nabla \times (\nabla \times \mathbf{C}) = \nabla(\nabla \cdot \mathbf{C}) - \nabla^2 \mathbf{C}$, to

obtain:

$$\begin{aligned}\frac{1}{b} \frac{\partial}{\partial \tau} \left[\frac{1}{b} \frac{\partial \mathbf{B}}{\partial \tau} \right] - \nabla^2 \mathbf{B} &= \frac{1}{b} [4\pi \nabla \times (\rho \mathbf{u})] \\ \frac{1}{b} \frac{\partial}{\partial \tau} \left[\frac{1}{b} \frac{\partial \mathbf{E}}{\partial \tau} + \frac{4\pi \rho \mathbf{u}}{b} \right] - \nabla^2 \mathbf{E} &= -[\nabla (4\pi \rho)].\end{aligned}$$

Using the fact that b is not constant, computing and rearranging terms, we obtain:

$$\begin{aligned}\frac{1}{b^2} \frac{\partial^2 \mathbf{B}}{\partial \tau^2} - \frac{\mathbf{u} \cdot \mathbf{a}}{b^4} \left[\frac{\partial \mathbf{B}}{\partial \tau} \right] - \nabla^2 \cdot \mathbf{B} &= \frac{1}{b} [4\pi \nabla \times (\rho \mathbf{u})], \\ \frac{1}{b^2} \frac{\partial^2 \mathbf{E}}{\partial \tau^2} - \frac{\mathbf{u} \cdot \mathbf{a}}{b^4} \left[\frac{\partial \mathbf{E}}{\partial \tau} \right] - \nabla^2 \cdot \mathbf{E} &= -\nabla (4\pi \rho) - \frac{1}{b} \frac{\partial}{\partial \tau} \left[\frac{4\pi (\rho \mathbf{u})}{b} \right],\end{aligned}\tag{3.23}$$

where $\mathbf{a} = d\mathbf{u}/d\tau$ is the acceleration vector. Thus, a new term arises when the proper time of the charge is used to describe its fields. This makes it clear that the local clock encodes information about the particle's interaction that is unavailable when the clock of the observer, co-moving observer or the proper clock of the center of mass is used to describe the fields. The new term in Equation (3.23) is dissipative, acts to oppose the acceleration, and is zero when $\mathbf{a} = 0$ or perpendicular to $\mathbf{u}$. This also occurs instantaneously with applied force. Furthermore, this term does not depend on the nature of the applied force. This is exactly what one expects of the back reaction caused by the inertial resistance of the particle to accelerated motion and, according to Wheeler and Feynman [75], is precisely what is meant by the radiation reaction.

Remark 15. *It is of particular interest that this implies that a charged particle can distinguish between inertial and accelerating frames. Thus, an observer in an elevator can always determine whether it is in a state of accelerated motion. From this point of view, it is no surprise that the 2.7 °K CMBR represents a uniquely preferred frame of rest.*

Let $(\mathbf{x}(\tau), \tau)$ represent the field position and $(\bar{\mathbf{x}}(\tau'), \tau')$ the retarded position of a source charge e , with $\mathbf{r} = \mathbf{x} - \bar{\mathbf{x}}$. If we set $r = |\mathbf{x} - \bar{\mathbf{x}}|$, $s = r - (\frac{\mathbf{r} \cdot \mathbf{u}}{b})$, and $\mathbf{r}_u = \mathbf{r} - \frac{r}{b} \mathbf{u}$, then in [39] the $\mathbf{E}$ and $\mathbf{B}$ fields were computed directly (using only retarded potentials), to obtain:

$$\mathbf{E}(\mathbf{x}, \tau) = \frac{q \mathbf{r}_u \left(1 - \mathbf{u}^2/b^2\right)}{s^3} + \frac{q [\mathbf{r} \times (\mathbf{r}_u \times \mathbf{a})]}{b^2 s^3} + \frac{q (\mathbf{u} \cdot \mathbf{a}) [\mathbf{r} \times (\mathbf{u} \times \mathbf{r})]}{b^4 s^3}$$

and

$$\mathbf{B} = \frac{q (\mathbf{r}_u \times \mathbf{r}) \left(1 - \mathbf{u}^2/b^2\right)}{r s^3} + \frac{q \mathbf{r} \times [\mathbf{r} \times (\mathbf{r}_u \times \mathbf{a})]}{r b^2 s^3} + \frac{q (\mathbf{u} \cdot \mathbf{a}) [\mathbf{r} \times \mathbf{u}]}{b^4 s^3}.$$

It is easy to see that $\mathbf{B}$ is orthogonal to $\mathbf{E}$. Except for τ in place of t , the first two terms in the above two equations are the same as in (19-13) and (19-14) in [60] (pg. 299). The last term in each case is new and arises because of

the dissipative terms in equations (3.23). These terms are zero if $\mathbf{a}$ is zero or is orthogonal to $\mathbf{u}$. In the first case, there is no radiation and the particle moves at a constant velocity such that the field is massless. The second case depends on the creation of motion that keeps $\mathbf{a}$ orthogonal to $\mathbf{u}$ (for example a cyclotron). Because $\mathbf{r} \times (\mathbf{u} \times \mathbf{r}) = r^2 \mathbf{u} - (\mathbf{u} \cdot \mathbf{r}) \mathbf{r}$, there is a component along the direction of propagation (longitudinal). (Thus, the $\mathbf{E}$ field has a longitudinal part.) This confirms our claim that the new dissipative term is equivalent to an effective mass for the particle. This implies that the radiation reaction comes directly from the use of the local clock to formulate Maxwell's equations. Thus, there is no need to assume advanced potentials, self-interaction or mass renormalization along with the Lorentz-Dirac equation to account for the radiation reaction as is required when the observer clock is used (Dirac or Wheeler-Feynman theory). Furthermore, no assumptions regarding a point particle or structure of the source are required (i.e., Poincaré stresses).

Remark 16. *We conjecture that this effective mass is the actual source of the photoelectric effect and that the photon is a real particle of non-zero (dynamical) mass, which travels with the fields but is not a part of the field in the normal sense. If this conjecture is correct, the radiation from a cyclotron (of any frequency) exposed to a metal like surface will not produce photo electrons.*

3.6.4. Global Internal Dynamics. In this section, we examine the motion of one particle, as seen from the global internal point of view. We assume that the system is closed or in equilibrium with a heat bath. In this case, we can assume that the Hamiltonian for each particle can be expressed as:

$$H_i = H_{i0} + V_i = \sqrt{c^2 \pi_i^2 + m_i^2 c^4} + V_i,$$

where $\pi_i = \mathbf{p}_i - \frac{e_i}{c} \mathbf{A}_i$, $\mathbf{A}_i = \sum_{j \neq i} \mathbf{A}_{ji}$ and $V_i = \sum_{j \neq i} V_{ji}$. We assume that $\mathbf{A}_{ji}, V_{ji}$ represents the retarded action of the vector potential (scalar potential) of the j -th particle on the i -th particle. Because we allow for n to vary and a charge at e_i may not equal the same value of e_j , we must allow for $e_i \mathbf{A}_{ji} \neq e_j \mathbf{A}_{ij}$ ($V_{ji} \neq V_{ij}$). We also do not include a customary factor of $1/2$ in our definition of the scalar and vector potentials acting on particle i (see [64]).

Recalling that $\mathbf{w}_i = d\mathbf{x}_i/dt$ and $\mathbf{u}_i = d\mathbf{x}_i/d\tau_i$, we define $\mathbf{v}_i = d\mathbf{x}_i/d\tau$. From our identities, we can easily see that:

$$\frac{\mathbf{w}_i}{c} = \frac{\mathbf{v}_i}{b} = \frac{\mathbf{u}_i}{b_i} \Rightarrow \gamma_i^{-1} = \sqrt{1 - \left(\frac{\mathbf{w}_i}{c}\right)^2} = \sqrt{1 - \left(\frac{\mathbf{v}_i}{b}\right)^2} = \sqrt{1 - \left(\frac{\mathbf{u}_i}{b_i}\right)^2}.$$

The velocity $\mathbf{v}_i$ is the one our observer sees when he uses the global canonical proper clock τ of the system to compute the particle velocity, whereas $\mathbf{w}_i$ is the one seen when he uses his clock to compute the particle velocity. If $\mathbf{U}$ is zero, $b = c$ and, from a global perspective, our theory resembles

the conventional theory. When the system is closed, $\mathbf{U}$ is constant and τ is linearly related to t . Because $\gamma_i^{-1} = \frac{1}{b} \sqrt{\mathbf{U}^2 + c^2 - \mathbf{v}_i^2}$, the physical interpretation is very different if $\mathbf{U}$ is not zero. Furthermore, it is easy to see that, even if $\mathbf{U}$ is zero in one frame, it will not be zero in any other frame which is in relative motion. Using $K = \frac{H^2}{2Mc^2} + \frac{Mc^2}{2}$, the equations of motion are as follows:

$$\begin{aligned} \mathbf{v}_i &= \frac{d\mathbf{x}_i}{d\tau} = \frac{\partial K}{\partial \mathbf{p}_i} = \frac{H}{Mc^2} \frac{c^2 \boldsymbol{\pi}_i}{H_{i0}} = \frac{b}{c} \frac{c^2 \boldsymbol{\pi}_i}{H_{i0}} = \frac{b}{c} \frac{d\mathbf{x}_i}{dt}, \\ \frac{d\mathbf{p}_i}{d\tau} &= \frac{\partial K}{\partial \mathbf{p}_i} = \frac{H}{Mc^2} \sum_{k=1}^n \left[\frac{c^2 \boldsymbol{\nabla}_i \boldsymbol{\pi}_k}{H_{i0}} - \boldsymbol{\nabla}_i V_k \right] = \frac{b}{c} \sum_{k=1}^n \left[\frac{c^2 \boldsymbol{\nabla}_i \boldsymbol{\pi}_k}{H_{i0}} - \boldsymbol{\nabla}_i V_k \right] = \frac{b}{c} \frac{d\mathbf{p}_i}{dt}. \end{aligned} \quad (3.24)$$

It is clear that the equations in (3.24) are mathematically equivalent. Factoring out the $k = i$ term $\frac{e_i}{b} [(\mathbf{v}_i \cdot \boldsymbol{\nabla}_i) \mathbf{A}_i + \mathbf{v}_i \times (\boldsymbol{\nabla}_i \times \mathbf{A}_i)]$, we have:

$$\begin{aligned} \frac{c}{b} \frac{d\mathbf{p}_i}{d\tau} &= \frac{e_i}{b} [(\mathbf{v}_i \cdot \boldsymbol{\nabla}_i) \mathbf{A}_i + \mathbf{v}_i \times (\boldsymbol{\nabla}_i \times \mathbf{A}_i)] \\ &+ \sum_{k \neq i}^n \left\{ \frac{e_k}{b} [(\mathbf{v}_k \cdot \boldsymbol{\nabla}_i) \mathbf{A}_k + \mathbf{v}_k \times (\boldsymbol{\nabla}_i \times \mathbf{A}_k)] - \boldsymbol{\nabla}_i V_k \right\}. \end{aligned} \quad (3.25)$$

Using

$$(\mathbf{v}_i \cdot \boldsymbol{\nabla}_i) \mathbf{A}_i = \frac{d\mathbf{A}_i}{d\tau} - \frac{\partial \mathbf{A}_i}{\partial \tau},$$

equation (3.25) becomes

$$\begin{aligned} \frac{c}{b} \frac{d\mathbf{p}_i}{d\tau} - \frac{e_i}{b} \frac{d\mathbf{A}_i}{d\tau} &= \frac{e_i}{b} [\mathbf{v}_i \times \mathbf{B}_i] - \frac{e_i}{b} \frac{\partial \mathbf{A}_i}{\partial \tau} - \boldsymbol{\nabla}_i V_i \\ &+ \sum_{k \neq i}^n \left\{ \frac{e_k}{b} [(\mathbf{v}_k \cdot \boldsymbol{\nabla}_i) \mathbf{A}_k + \mathbf{v}_k \times (\boldsymbol{\nabla}_i \times \mathbf{A}_k)] - \boldsymbol{\nabla}_i V_k \right\}. \end{aligned} \quad (3.26)$$

Note that equation (3.26) can also be written as:

$$\begin{aligned} \frac{d\mathbf{p}_i}{dt} - \frac{e_i}{c} \frac{d\mathbf{A}_i}{dt} &= \frac{e_i}{c} [\mathbf{w}_i \times \mathbf{B}_i] - \frac{e_i}{c} \frac{\partial \mathbf{A}_i}{\partial t} - \boldsymbol{\nabla}_i V_i \\ &+ \sum_{k \neq i}^n \left\{ \frac{e_k}{c} [(\mathbf{w}_k \cdot \boldsymbol{\nabla}_i) \mathbf{A}_k + \mathbf{w}_k \times (\boldsymbol{\nabla}_i \times \mathbf{A}_k)] - \boldsymbol{\nabla}_i V_k \right\}. \end{aligned} \quad (3.27)$$

Thus, Eqs. (3.26) and (3.27) are mathematically equivalent. Set $V_i = e_i \Phi_i$ and $\mathbf{E}_i = -\frac{1}{b} (\partial \mathbf{A}_i / \partial \tau) - \boldsymbol{\nabla}_i \Phi_i$, then we can write:

$$\mathbf{F}_i = \frac{e_i}{b} (\mathbf{v}_i \times \mathbf{B}_i) - \frac{e_i}{b} \frac{\partial \mathbf{A}_i}{\partial \tau} - \boldsymbol{\nabla}_i V_i = e_i \mathbf{E}_i + \frac{e_i}{b} (\mathbf{v}_i \times \mathbf{B}_i).$$

We can then write equation (3.26) as:

$$\begin{aligned} \frac{c}{b} \frac{d\pi_i}{d\tau} &= \mathbf{F}_i \\ &- \sum_{k \neq i}^n \left\{ \frac{e_k}{b} [(\mathbf{v}_k \cdot \nabla_k) \mathbf{A}_{ik} + \mathbf{v}_k \times (\mathbf{v}_k \times \mathbf{A}_{ik})] - \nabla_k V_{ik} \right\}. \end{aligned}$$

If we now use

$$\begin{aligned} (\mathbf{v}_k \cdot \nabla_k) \mathbf{A}_{ik} &= \frac{d\mathbf{A}_{ik}}{d\tau} - \frac{\partial \mathbf{A}_{ik}}{\partial \tau}, \quad \mathbf{B}_{ik} = \nabla_k \times \mathbf{A}_{ik}, \\ \mathbf{E}_{ik} &= -\frac{1}{b} \frac{\partial \mathbf{A}_{ik}}{\partial \tau} - \nabla_k \Phi_{ik}, \quad \mathbf{F}_{ik} = e_k \mathbf{E}_{ik} + \frac{e_k}{b} (\mathbf{v}_k \times \mathbf{B}_{ik}), \end{aligned}$$

the above becomes:

$$\frac{c}{b} \frac{d\pi_i}{d\tau} = \mathbf{F}_i - \sum_{k \neq i}^n \left[\mathbf{F}_{ik} - \frac{e_k}{b} \frac{d\mathbf{A}_{ik}}{d\tau} \right]. \quad (3.28)$$

If we simplify and put the last term on the other side, we have:

$$\frac{c}{b} \sum_{i=1}^n \frac{d\pi_i}{d\tau} + \sum_{i=1}^n \sum_{k \neq i}^n \frac{e_k}{b} \frac{d\mathbf{A}_{ik}}{d\tau} = \sum_{i=1}^n \mathbf{F}_i - \sum_{i=1}^n \sum_{k \neq i}^n \mathbf{F}_{ik}.$$

Performing the summations on both sides give us:

$$\begin{aligned} \frac{c}{b} \sum_{i=1}^n \frac{d\pi_i}{d\tau} + \sum_{i=1}^n \frac{e_k}{b} \frac{d\mathbf{A}_{ik}}{d\tau} &= 0 \Rightarrow \\ \sum_{i=1}^n \frac{d\mathbf{p}_i}{d\tau} &= 0 = \frac{d\mathbf{P}}{d\tau}. \end{aligned}$$

3.6.5. Discussion. In this section, we first discuss the relationship between equations (3.18) and (3.28). For comparison, we first rewrite Equation (3.18) using its indices.

$$\frac{c}{b_i} \frac{d\pi_i}{d\tau} = \left[e_i \mathbf{E}_i + \frac{e_i}{b_i} (\mathbf{u}_i \times \mathbf{B}_i) \right] = \frac{d\pi_i}{dt}. \quad (3.29)$$

If we use

$$\mathbf{F}_i = e_i \mathbf{E}_i + \frac{e_i}{b} (\mathbf{v}_i \times \mathbf{B}_i).$$

We can write equation (3.28) as

$$\frac{d\pi_i}{dt} = \frac{c}{b} \frac{d\pi_i}{d\tau} = \left[e_i \mathbf{E}_i + \frac{e_i}{b} (\mathbf{v}_i \times \mathbf{B}_i) \right] - \sum_{k \neq i}^n \left[\mathbf{F}_{ik} - \frac{e_k}{b} \frac{d\mathbf{A}_{ik}}{d\tau} \right]. \quad (3.30)$$

Equation (3.29) represents one particle in the force field, as seen locally. It does not react via the action-at-a-distance, and its reaction appears in its field via the additional term in its wave equation. When we look at the same particle from the center-of-mass frame (equation (3.30)), we see the

force acting on the particle and the delayed action-at-a-distance reaction force of the particle on each of the other particles in the system.

Thus, the extra term on the (far) right-hand side of equation (3.30) is the back-reaction field of the i -th particle on each of the other particles in the system. This term accounts for the radiation reaction as action at a distance. There is no self-energy (divergence), advanced potentials, or any assumptions regarding the size or structure of the particles. It is important to note that the mathematical equivalence is manifested in both cases (using t or τ for the center of mass). However, τ_i must be use at the local level (see Theorem 3.1).

It is clear that equation (3.30) is consistent with conservation of the global momentum. This along with the conservation of total energy, implies the following.

Theorem 17. (Wheeler-Feynman) *If we use the τ_i at the local level with either the $(\mathbf{X}, t)$ or $(\mathbf{X}, \tau)$ variables, the closed system of interacting charged particles exchange energy and momentum via fields at the local level and by action-at-a-distance as seen from the global internal view. In addition, at the global level, all emitted energy and momentum are absorbed internally.*

Thus, for any closed system of interacting particles, the Wheeler and Feynman absorption hypothesis is automatically satisfied without the use of advanced potentials.

Remark 18. *Wheeler and Feynman [75] conjectured that action-at-a-distance and (classical) field theory represent different sides of the same theory. The discussion above explains why and how their conjectures are true. At the local level the individual particle is immersed in an electromagnetic field, whereas an observer at the center of mass sees the particle being acted on by all the other particles and the corresponding direct (delayed) back action by the particle on each of the others. This suggests that, because Maxwell's equations are spin 1 relativistic wave equations, the appearance of the 2.7 °K CMBR is a natural consequence of delayed interaction, allowing some photons to miss their intended mark (violating Newton's third law). This would also help to explain why the probability interpretation in quantum mechanics works well.*

3.7. Dual Lorentz Group. We now identify the dual Lorentz transformation group which preserves the first postulate of the special theory. The standard (Lorentz) time transformations between two inertial observers for the i^{th} particle can be written as

$$t' = \gamma(\mathbf{v}_i) [t - \mathbf{x}_i \cdot \mathbf{v}_i / c^2], \quad t = \gamma(\mathbf{v}_i) [t' + \mathbf{x}'_i \cdot \mathbf{v}_i / c^2]. \quad (3.31)$$

We want to replace t , t' by τ_i . To do this, use the relationship between dt and $d\tau_i$ to get:

$$t = \frac{1}{c} \int_0^{\tau_i} b_i(s) ds = \frac{1}{c} \bar{b}_i \tau_i, \quad t' = \frac{1}{c} \int_0^{\tau_i} b'_i(s) ds = \frac{1}{c} \bar{b}'_i \tau_i, \quad i = 1, \dots, n. \quad (3.32)$$

where we have used the mean value theorem of calculus to obtain the end result, so that both $\bar{b}_i$ and $\bar{b}'_i$ represent an earlier τ_i -value of b_i and b'_i respectively. Note that, as b and b' depend on τ , the transformations (3.32) represent explicit nonlocal, nonlinear relationships between t , t' and τ_i (during interaction). (This is to be expected in the general.) Suppressing the index, if we set

$$\mathbf{d}^* = \mathbf{d}/\gamma(\mathbf{v}) - (1 - \gamma(\mathbf{v})) [(\mathbf{v} \cdot \mathbf{d})/(\gamma(\mathbf{v})\mathbf{v}^2)] \mathbf{v},$$

we can write the transformations that fix τ as:

$$\begin{aligned} \mathbf{x}' &= \gamma(\mathbf{v}) [\mathbf{x}^* - (\mathbf{v}/c)\bar{b}\tau], & \mathbf{x} &= \gamma(\mathbf{v}) [\mathbf{x}'^* + (\mathbf{v}/c)\bar{b}'\tau], \\ \mathbf{u}' &= \gamma(\mathbf{v}) [\mathbf{u}^* - (\mathbf{v}/c)b], & \mathbf{u} &= \gamma(\mathbf{v}) [\mathbf{u}'^* + (\mathbf{v}/c)b'], \\ \mathbf{a}' &= \gamma(\mathbf{v}) \{\mathbf{a}^* - \mathbf{v}[(\mathbf{u} \cdot \mathbf{a})/(bc)]\}, & \mathbf{a} &= \gamma(\mathbf{v}) \{\mathbf{a}'^* + \mathbf{v}[(\mathbf{u}' \cdot \mathbf{a}')/(b'c)]\}. \end{aligned} \quad (3.33)$$

If we put equation (3.32) in (3.31), differentiate with respect to τ and cancel the extra factor of c , we get the transformations between b and b' :

$$b'(\tau) = \gamma(\mathbf{v}) [b(\tau) - \mathbf{u} \cdot \mathbf{v}/c], \quad b(\tau) = \gamma(\mathbf{v}) [b'(\tau) + \mathbf{u}' \cdot \mathbf{v}/c]. \quad (3.34)$$

Equations (3.33) in (3.34) provide an explicit representation of transformation group, which uses the particle proper time to describe the dynamics and preserves the first postulate of the special theory. We call this group the dual Lorentz group.

It was shown in [40] that Maxwell's equations transform in the same way as in the conventional theory. However, the charge and current density have the following transformations:

$$\mathbf{J}' = \mathbf{J} + (\gamma - 1) \frac{\mathbf{J} \cdot \mathbf{v}}{\mathbf{v}^2} \mathbf{v} - \frac{\gamma b}{c} \rho \mathbf{v}, \quad (3.35)$$

$$b' \rho' = \gamma(\mathbf{v}) [b\rho - (\mathbf{J} \cdot \mathbf{v}/c)]. \quad (3.36)$$

Using the first equation of (3.34) in (3.36), we get:

$$\rho' = \frac{\rho - (\mathbf{J} \cdot \mathbf{v}/bc)}{1 - (\mathbf{u} \cdot \mathbf{v}/bc)}. \quad (3.37)$$

This result is different from the standard one, (which we obtain if we set $b' = b = c$ in (3.36),

$$\rho' = \gamma(\mathbf{v}) [\rho - (\mathbf{J} \cdot \mathbf{v}/c^2)].$$

Furthermore, if we insert the expression $\mathbf{J}/c = \rho(\mathbf{u}/b)$ in (3.36); we obtain

$$\rho' = \rho \frac{1 - (\mathbf{u} \cdot \mathbf{v}/b^2)}{1 - (\mathbf{u} \cdot \mathbf{v}/bc)}. \quad (3.38)$$

In order to see the impact of equation (3.38), suppose that a spherical charge distribution is at rest in the unprimed frame. From (3.38), we see that $\mathbf{u} = 0$, so that $\rho' = \rho$. Since the primed frame is arbitrary, the charge distribution will appear spherical to all observers. This is what we would expect on physical grounds, so that relatively moving frames should not change the symmetry properties of charged objects. In particular, a spherical charge

distribution in one frame should not be distorted in another moving frame (i.e., display physical effects due to another observer's relative motion).

3.8. Mathematical Equivalence. It may be unsettling to some that two theories could be mathematically equivalent but not physically equivalent. It is more natural to expect that two mathematically equivalent theories would also be physically equivalent, and there are a number of historical examples to support such expectations; the Lagrange-Hamiltonian formulation of classical mechanics, the Heisenberg-Schrödinger formulation of quantum mechanics, and the Feynman-Schwinger-Tomonaga formulation of quantum electrodynamics. In the first case, both formulations have proved equally valuable depending on the purpose. However, the latter two cases raise interesting questions. After Feynman constructed a path integral formulation of quantum mechanics, it was shown to physically include the Heisenberg-Schrodinger formulation. However, it was only recently shown to be mathematically equivalent, and a new Hilbert space had to be designed for that purpose (see [33]).

In our opinion, the Feynman path integral approach is physically more general than that of Heisenberg and Schrödinger, and his formulation of QED is physically more general than that of Schwinger and Tomonaga. In both cases, he introduces new concepts that make it physically easier to think about and solve problems. What Feynman did was to show that it is still possible to formulate theories which more closely represent the way the world appears to us in our consciousness.

It was Poincaré [59], who first noticed that some hypotheses (or assumptions), which are made for theory construction, arise because of empirical data, while others occur because they are convenient. The convenient hypotheses are generally imposed by the mathematical structures we use to represent physical theories. These hypotheses are called conventions by Poincaré in order to point out the fact that different conventions could lead to different theories which would be mathematically equivalent. He was not sure that the theories would be physically different, but he seems to have left open that possibility.

4. LIGHT: ITS NATURE, ITS MASS AND ITS SPEED

4.1. Photon Nature. Given that, in our local-time formulation of the Maxwell theory, the value of the “natural” speed of light depends on the motion of the source, it is not presumptuous to take another look at the age-old problem of its nature and seek to understand the impediments to treating a photon on the same footing as any other elementary particle.

The single most important problem with any attempt to treat photons as elementary particles is its well-known wave property. Interference and diffraction experiments and, indeed, a number of major fields of electrical engineering attest to the amazing precision, effectiveness and efficiency of

the wave picture. The field theory view of both photons and particles is that they are localized wave packets of the quantized electromagnetic field (or the particular particle field). Hence, one expects that matter will behave like particles in some experiments and like waves in others and, as such, may be viewed as some type of atomic compromise between two distinct classical views. However, these ideas go back to the fundamental work of Born, deBroglie and Heisenberg, while today we have much more experimental information about photons since those pioneering efforts.

It has now been known for over 35 years that we can control the intensity of a beam of photons in interference (or diffraction) experiments to the point that individual photons may be counted on a photographic plate (see Paul [57]). Furthermore, the distribution of photons appears random and, only after a long time period (depending on the intensity level), do we begin to see wave patterns. These experiments, along with the photoelectric and Compton effects, conclusively tell us the following:

- (1) It is not unreasonable to treat photons as elementary particles.
- (2) The wave length and frequency are characteristic of groups of photons.
- (3) Concepts such as electric and magnetic fields are macro-properties that have some (but limited) reality on the atomic scale and (possibly) none at the sub-atomic scale.

For an excellent discussion of the above, see Buenker [13].

4.2. Photon Mass. In the past, work on the question of photon mass focused on the addition of a mass term to the Lagrangian density for Maxwell's equations and generally leads to the Proca equation (see Bargmann and Wigner [14]). Early work in this direction can be traced back from the paper of Schrödinger and Bass [70]. As in our approach, the speed of light is no longer constant in all reference frames. In this case, the fields are distorted by the mass term and experiments of Goldhaber and Nieto [30] use geomagnetic data to set an upper bound of $3 \times 10^{-24} \text{ GeV}$ for the mass term (see Jackiw [45]). This approach causes gauge formulation difficulties, and has not found favor at the classical level.

It should be recalled that Maxwell's equations are (spin 1) relativistic wave equations (see [14]). On the other hand, experiments show directly that photons have a weight as one would expect of any material object. These experiments do not depend on either the special or general theory of relativity (according to Pound and Snider [61]) and are not directly dependent on frequency or wavelength measurements.

In a recent paper, Afshar et. al. [2] report on the presence of sharp interference and highly reliable which-way information in the same experimental arrangement for the same photons using new non-perturbative measurement techniques at separate spacetime coordinates, both of which refer back to

the behavior of the photon at the same event (i.e., the passage through the pinholes).

They inferred full fringe visibility from the observation that the total photon flux was only slightly decreased when thin wires were placed exactly at the minima of the interference pattern. Which-way information was then obtained downstream via imaging using a special lens system. This allowed them to determine which-way information about each photon as it passed the plane of the pinholes. The experiment allowed them to circumvent the predicted limitations imposed by the uncertainty principle and the entanglement between the which-way marker and the interfering quantum object as employed in other experiments. They further point out that the non-perturbative measurement technique used in the experiment is conceptually quite different from quantum non-demolition or non-destructive techniques which perturb the photon wave function directly. This shows that wave-particle duality is more a property of experimental approach and technique as opposed to an intrinsic property of photons.

It is not unreasonable to correlate the frequency of a photon with that of linear harmonic oscillations for low energy fields. Drozov and Stahlhofen [21] note that the Planck formula, $\mathcal{E} = h\nu$, was based on the fact that the energy emitted from atoms resulted from electromagnetic waves propagating as continuous harmonic oscillations. In order to solve the cavity problem, Planck postulated energy quantization in terms of the frequencies defined by the oscillations, leading to the above formula, where the total energy of the emitted electromagnetic wave is an integer multiple of h . From the success of this approach, it is now natural to believe that photons are small parts of the electromagnetic field. However, such a view is untenable for describing the ultra-short pulses that are obtained in laboratories today. If we combine the Planck formula with $c = \lambda\nu$, one must then accept that $\lambda = hc/\mathcal{E}$. Since we now don't have wave-particle dualism to save us, the concept of a photon as a wave begs the question posed by Drozov and Stahlhofen: "how many oscillations does it contain?". We agree with them that it is much more reasonable to relate ν to the source emission time. From this view, the emission time is only restricted by the time energy uncertainty relationship $\Delta\mathcal{E}\tau \approx h/2$. Thus, we conclude that there is no oscillation period with frequency ν , just the characteristic time $\tau = 1/(2\nu)$, for a single (massive) energy pulse. (We note that the use of a Fourier series representation, as is generally done in optics, is no more than a good mathematical device and should not be used for physical interpretation.) In closing, we point out that, as noted by Feynman [29], a small photon mass will eliminate the infrared catastrophe in QED.

4.3. Light Speed. In this and the remaining sections, we give additional consideration to the physical implications of our interpretation of b and b' as the speed of light relative to the source for the different observers (collaborative speed of light). We also consider a few topics in astrophysics

in which the conventional theory requires additional assumptions in order to explain observations that point to speeds higher than c .

In order to gain some perspective, suppose an emitting system is at rest in the unprimed frame so that $b = c$. In this case, the collaborative speed of light observed in the primed frame is $b' = \gamma(\mathbf{v})c$ and the velocity of the source is seen as $\mathbf{u}' = -\gamma(\mathbf{v})\mathbf{v}$. Thus, if the two observers are separating at high speeds, both b' and $\mathbf{u}'$ may be very large.

There are some experiments where use of the observer's clock provides a clear answer. A classic example is the Michelson-Morley experiment. This experiment gave the first bell of doom for the ether theory, and is easily explained by the special theory (using the clock of the observer). It also has a simple explanation when the clock of the source is used since, in this case, the source is at rest in the frame of the observer so that $\mathbf{u} = \mathbf{0} \Rightarrow b = c$.

It is clear that, at the speeds obtained in the world of our ordinary experience, no significant difference between the two approaches is expected. However, at high energies and/or small distances, we expect differences to show up in a dramatic way. Indeed they have, but the definition of velocity depends on the clock attached to the observer, $\mathbf{w} = d\mathbf{x}/dt$, while all contrary results are interpreted as due to time dilation. For example, without a switch in clocks, the existence of cosmic ray muons at the surface of the earth have no explanation.

An equally valid (and physically consistent) interpretation is that the velocity of the particle is not $\mathbf{w}$, but $\mathbf{u} = d\mathbf{x}/d\tau$ and, in this case, no contrary results occur. The use of $\mathbf{w}$ is clearly a convenient choice for most of ordinary physics (where both choices are the same). However, in high-energy experiments, the local clock of the system is necessary (and used) to analyze and explain scattering events by using time dilation to make the results correspond to velocities below c .

In order to obtain a different view of experiments on the lifetime of fast mesons and the velocity of γ rays and light from moving sources, first consider the definition of momentum. When the clock of the observer is used to measure time, momentum increase is attributed to relativistic mass increase so that $\mathbf{p} = m\mathbf{w}$ and $m = m_0[1 - w^2/c^2]^{-1/2}$. On the other hand, if we use the clock of the source, we have $\mathbf{p} = m_0\mathbf{u}$ and $\mathbf{u} = \mathbf{w}[1 - w^2/c^2]^{-1/2}$ so that there is no mass increase, the (proper) velocity increases. Thus, in particle experiments, the particle has a fixed mass and invariant decay constant, independent of its velocity, but can have speeds $> c$. An analysis of experiments on the lifetime of fast mesons, the velocity of γ rays and light from moving sources reveal that, at some point, either the speed of light is assumed to be independent of the motion of the source, or time dilation is used. Both assumptions imply that only the clock of the observer is used. This is basically inconsistent since one is using two clocks to explain one experiment. Either, the speed of the particle is less than c as determined by the observer's measuring instruments and clock, or, if time dilation is used,

one is in fact inferring that the clock of the particle is needed to explain the experiment. In the latter case, the actual speed of the particle is then $\mathbf{u}$, which can be greater than c . The problem is that one cannot use two clocks and claim consistency with the basic measurement framework not for just the special relativity, but all of science. *Thus, the analysis of these experiments is flawed in its validation of the conventional theory and does not prove that the speed of light relative to the particle is c .* (Furthermore, most particles of interest are not in an inertial frame, so the second postulate does not apply.)

4.4. Relativistic Jets in Our Galaxy. In 1918 Curtis [17] made the first discovery of jet-like features emanating from the nuclei of galaxies. He identified a jet in the optical range from an elliptical galaxy in the Virgo cluster (M87). Since then, a large number of objects with a jet-like structure have been discovered. However, starting about 30 years ago, researchers began to find quasars with jet expansions of up to ten or more times the speed of light (see Pearson and Zensus [65]; [77] Zensus; Mirabel and Rodríguez [51]). This started quite a stir at the time and led many to suggest the possible breakdown of the special theory. Since the source appeared to be at rest relative to earth, unlike the decay of fast muons from the top of the atmosphere, the assumption of time-dilation would not work. Rees has suggested that we may be looking at the jets from some angle relative to the observer. He has shown how these large speeds may be an aberration because we were looking at a projection of the true image onto the plane of view of the observer (see [66]). This would explain the apparent approaching and receding condensations with very different velocities. However, this is an additional assumption has no independent verification and has not been completely accepted (see Mirabel and Rodríguez [51]; De Rújula [68]).

With the conventional definition of velocity, it is very difficult to imagine even the most powerful astrophysical systems, such as active galactic nuclei and/or radio galaxies, accelerating heavy nuclei or protons to the required high energies within existing physical theories. However, if the local clock of the ejecting system determines the distance traveled and speed ($\mathbf{u}$) of a particle, then no new particles or the breakdown of the special theory are needed to explain the results. Since most measurements (physical and astronomical) are based on the dimensionless ratio $\beta = \mathbf{w}/c \equiv \mathbf{u}/b$, we see that the results will not change. For example, ([55], pp. 556-561), the red shift factor z , used to determined distances in astronomy, is defined by:

$$z = \sqrt{\frac{1 + \frac{\mathbf{w}}{c}}{1 - \frac{\mathbf{w}}{c}}} - 1 \equiv \sqrt{\frac{1 + \frac{\mathbf{u}}{b}}{1 - \frac{\mathbf{u}}{b}}} - 1.$$

(The formula $z = \frac{\mathbf{w}}{c} = \frac{\mathbf{u}}{b}$ is normally used for values of $|\mathbf{w}| \leq 10^4 km/sec$.) We thus conclude that light may reach us from much farther away and with more intensity than is traditionally expected. Thus, we may well be looking

at some galaxies that are not as close and others that are not as far as predicted from conventional theory.

5. Historical Time, CMBR and the Symmetric Big Bang

Let M be the effective mass of a system of particles defined on phase space with variables $(t, \mathbf{X}, \mathbf{P})$ as seen by observer O , and variables $(t', \mathbf{X}', \mathbf{P}')$ as seen by observer O' (both in inertial frames). If $\gamma^{-1}(\mathbf{V}) = \sqrt{1 - \frac{\mathbf{V}^2}{c^2}}$, $\mathbf{V} = \frac{d\mathbf{X}}{dt}$, the relations $H = \gamma(\mathbf{V}) Mc^2$ for observer O and $H' = \gamma(\mathbf{V}') Mc^2$ for observer O' , allow us to also represent the proper time as:

$$d\tau = \frac{Mc^2}{H} dt, \quad d\tau = \frac{Mc^2}{H'} dt'. \quad (5.1)$$

5.1. HF-Time and the CMBR. Newton believed that absolute time exists independent of any observer, and provides a universal definition of simultaneity and ordering for events, but can only be understood mathematically. (In Newton's era mathematics and personal philosophy still played a dominant role in one's physics). Einstein's special theory appeared to eliminate Newton's absolute time and Minkowski's additional postulate eliminated the ordering property. Feynman reintroduced the ordering property by suggesting that we view a physical event as a 3-dimensional motion picture, exposing more of the outcome as the film unfolds.

The use of a fifth parameter to provide the missing order property of time began with Fock, then Stueckelberg, followed by Feynman and Schwinger. They considered it a global clock for Minkowski space time. Horwitz and Piron [44], and later Fanchi and Collins called it the historical time. Horwitz and Fanchi were the first to realize that this clock should have physical justification (see [27]). In this section, we provide two theorems for the existence of historical time. The first proof is based on equation (5.1) and a weak assumption about the properties of the universe, compared to the cosmological principle, and the proof was given in Foundations 1 [35]. It was motivated by the studies of Horwitz and Fanchi (see [40] and [34]). The second proof is based on the existence of the 2.7 °K CMBR.

Theorem 19. *If the universe is representable in the sense that it is independent of the observer's portion of the universe, then the universe has a unique clock available to all observers.*

In 1965, Penzias and Wilson discovered a 2.7 °K CMBR. Peebles suggested that, the special theory is valid with or without a preferred frame, so that the CMBR does not violate the special theory. General relativity predicts, that at each point, one can adjust their acceleration locally to find a freely falling frame where the special theory holds. In this frame, the observers with constant velocities are equivalent. Thus, according to the general theory there is an infinite family of freely falling frames. The findings of Penzias and Wilson showed that the acceleration can be set equal to

zero. Additional historical information and detail can be found in Peebles [54].

During the Einstein Centennial Symposium, held in 1979 at the University of Illinois at Carbondale, Dirac noted that the CMBR dipole anisotropy makes it possible to identify the relative velocity and direction of the frame where the CMBR is at rest. Measurements at the time showed that this frame is directed towards the Hydra-Centaur with a speed of 627km/sec. Dirac then suggested that this corresponds to the reference frame where the Big-Bang occurred, making it a privileged frame that contradicts the general theory of relativity. During his talk Wigner noted that general relativity only assures the equivalence of the laws of nature in each reference frame and makes no claims about the initial conditions for the universe; thus, a reference frame at rest with the Big-Bang does not contradict the general theory of relativity (see [22]).

We now know that this radiation represents a unique preferred frame of rest that exists throughout the universe and is available to all observers (see [62]). This radiation is highly isotropic with anisotropy limits set at 0.001%. Furthermore, direct measurements were made of the velocity of our Solar System and Galaxy (370 and 600 *km/sec* respectively). de Parga et. al. have shown that if a detector is placed in the direction of the Hydra-Centaur with a speed of 627km/sec, the particle number density will be independent of the angle, so that the dipolar anisotropy, which is a directional temperature will disappear (see [3]).

The following Poincaré's analysis of hypotheses in physics, we see that the following convenient assumption about the CMBR, which is also based on empirical data, is as much important to our understanding of the universe as it's support for the current version of the big bang.

Theorem 20. *If all observers choose a frame at rest with respect to the $2.7^\circ K$ cosmic microwave background radiation, then their clocks will run on historical time and the laws of physics will be the same for all observers.*

Proof. First, the CMBR fills all space; therefore it is available to all observers. Second, any observer can detect their motion relative to the CMBR. Thus, each observer can transform into a CMBR frame. Because this frame is the same at every point in the universe, all clocks will provide the same measurement of time. $\square$

Remark 21. *In our opinion, the most important aspect of the CMBR is it has a blackbody spectrum. It was the analysis and understanding of this spectrum by Planck that led to the discovery of quantum mechanics. To understand the implications of this observation, we first need to review the experimental conditions that were required to reveal the spectrum in the laboratory:*

- (1) *A closed container except for a small hole for the measurements was required.*

- (2) *The ability to increase the internal temperature from the outside was required.*

These conditions for the observation of such a spectrum leads us to the following conclusions about the universe:

- (1) *The universe is closed.*
 (2) *Energy from outside the universe is required.*
 (3) *Quantum mechanics is operative at the cosmic scale.*

5.2. The Iso-Dual Numbers. As noted in the first of this series [33] the use of a global variable without attaching physical meaning to it dates back to the early work of Tetrode and Fock (for a review, see Fanchi [26]). However, starting in the 1970's, Horwitz and Piron and later Fanchi began to suggest the use of a special clock for global systems which they called the historical time (see [27]). They predicted that such a variable should exist as a real physical parameter and Fanchi suggested experiments to detect this clock.

One might naturally suspect that Dirac would have noticed that the positron could be interpreted as an electron moving backward in time after developing his hole theory. However, according to Schweber (see pg. 38-388 [73]), in 1940 Wheeler suggested to Feynman that one might interpret a positron as an electron moving backwards in proper time, with positive charge. He also points out that Wheeler was unaware of Stueckelberg's later work on the same subject in a 1942 paper (see [72]).

The correct mathematical system for such an interpretation only became possible after Santilli developed his theory of iso-dual numbers (see [69]). Let's revisit our conceptual view of the real numbers and their representation. Recall that mathematically, a field is any set $\mathbb{F}$ that has two binary operations $\oplus$ and $\odot$ that satisfies all our common experience with real numbers.

Example 22. *For example let $\mathbb{F}$ be the set of all diagonal $n \times n$ matrices.*

Definition 23. *The real numbers are a quadruple $(\mathbb{R}, +, \cdot, \rightarrow)$, which is a field, with 0 as the additive identity and 1 as the multiplicative identity.*

We have included $\rightarrow$ in order to include the unacknowledged sense of direction. If $\hat{\mathbb{R}} = \{-a : a \in \mathbb{R}\}$, then:

Definition 24. *The dual of the real numbers $(\hat{\mathbb{R}}, +, * \leftarrow)$ is a field, with $0 = \hat{0}$ as the additive identity, $-1 = \hat{1}$ as the multiplicative identity and,*

$$\hat{a} * \hat{b} = (-a)(-1)(-b) = (-ab) = \hat{a}\hat{b} \text{ for all } \hat{a}, \hat{b} \in \hat{\mathbb{R}}.$$

We note that we can obtain the dual of any physical quantity $\hat{A}$ from the equation $A + \hat{A} = 0$.

Example 25. *A simple example from quantum theory is as follows: the evolution of a particle is defined on a Hilbert space $\mathcal{H}$ over complex numbers $\mathbb{C} = \mathbb{R} + i\mathbb{R}$, with Hamiltonian H given by the equation:*

$$i\hbar \frac{\partial \psi}{\partial \tau} = K\psi.$$

The conjugate equation is:

$$-i\hbar \frac{\partial \psi^*}{\partial \tau} = K\psi^*.$$

If we use $\hat{\mathbb{C}}$ as our number field, we can write the above equation as:

$$\hat{i} * \hat{\hbar} * \frac{\partial \hat{\psi}^*}{\partial \hat{\tau}} = \hat{K} * \hat{\psi}^*$$

This will be used in the third paper of this series to show that for the dual quantum theory it is natural to view an antiparticle as a particle with its proper time reversed, and with its evolution defined on $\mathcal{H}^$ over $\hat{\mathbb{C}}$.*

Remark 26. *Santilli has shown in [69] that charge conjugation and isoduality are both equivalent for the particle-antiparticle symmetry operation.*

5.3. The Symmetric Big Bang. In this section, we bring together the ideas of Feynman and Santilli to introduce a symmetric version of the big bang that has many advantages compared to the current model. The currently accepted model for the universe assumes that it began around 13.8 billion years ago with a big bang. It assumes that the laws of physics did not exist until after the first few Planck seconds. This model is problematic because of the following (incomplete list of) problems.

- (1) The model does not allow us to ask what happen before the big bang.
- (2) The model requires the suspension of all known laws of physics before the first few Planck seconds.
- (3) The model requires that we believe in something that has no basis in our reality.
- (4) To accept this model as final would leave the foundations of cosmology forever outside of science.

However, we first began observing creation events in the laboratory when Carl D. Anderson first discovered the electron-positron creation event in 1932, and there is no credible reason to accept the above problems as forever required (see [15]).

We make the fundamental assumption that, “the laws of physics have always existed.” The experimental conditions required for an elementary creation event in the laboratory are as follows:

- (1) Such an event requires a large outside light energy source containing energy in excess of twice the mass energy of one particle.

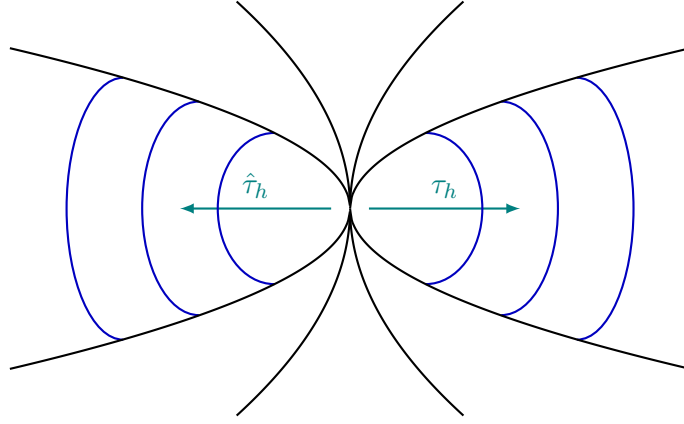


FIGURE 1. Symmetric Big Bang

- (2) Such an event requires a very large near by mass to absorb the reaction momentum.
- (3) The complete system conserves all (known) laws of physics.

We suggest a symmetric big bang beginning in the diagram below. In this view, two universes are created, one going forward in τ_h and the other going backward in τ_h (historical time), relative to our reality (Figure 1).

This model has the following advantages:

- (1) This model is compatible with creation events as obtained in the laboratory.
- (2) This model explains the lack of antimatter in the universe.
- (3) This model provides an arrow for historical time with a zero initial point.
- (4) This model requires an unknown, very large mass outside the universe to absorb the extremely large reaction momentum.
- (5) The model maintains the conservation of all known laws of physics (no proton decay).
- (6) The historical time is a proper time, so that the speed of the creation event is not bounded by c . Thus, inflation is not required to account for the horizon or the flatness problem and the second postulate is not violated.

6. Conclusion

This paper provided the classical foundations for quantum electrodynamics. In particular we have:

- (1) Provided the first relativistic unification of the Newtonian and Maxwell theories.

- (2) Resolve all problems associated with radiation reaction, self-energy divergence, field-particle dichotomy and advanced interactions in CED.

Recall that the self-energy divergence in QED came from CED and that Feynman observed that if the photon has a small mass, the infrared divergence in QED disappears. It follows that a second quantized version of the Einstein or Einstein dual theory will not have a self-energy or infrared divergence. This along with the solution to Dyson's conjectures given in [35], provides both understanding and correction for all the physical and mathematical problems associated with the divergencies in QED.

In the last section we provided another proof for the existence of a universal clock based on the existence of the 2.7 °K CMBR. (The first proof was proven in Foundations I [35], based on an assumption weaker than the cosmological principle.) We have also shown that a major implication of the CMBR is that the universe is closed and that quantum mechanics operates on the cosmological scale. In addition, we have used the historical time to provide a symmetric model for the big bang beginning, which corresponds to creation events seen in the laboratory. This approach has the following advantages:

- (1) This model is compatible with creation events obtained in the laboratory.
- (2) This model requires an outside light (energy) source.
- (3) This model explains the lack of antimatter in this universe.
- (4) This model provides an arrow for historical time with a zero initial point.
- (5) This model requires an unknown, very large mass outside the universe to absorb a large reaction momentum.
- (6) The model maintains the conservation of charge, energy, linear and angular momentum.
- (7) Baryon number is conserved (no proton decay).
- (8) Because historical time is proper the speed of the creation event is not bounded by c , so that inflation assumption is not required and the second postulate is not violated.
- (9) Antiparticles (in this universe) may naturally be viewed as particles moving backward in their proper time.

PROPER TIME HISTORY

The first time that proper time was used in the context of this work was by Hans Montanus, with a number of papers starting in 1991 (see [43] and references). He used the proper time of the observer as a fourth coordinate to develop a version of Maxwell's theory and the general theory of relativity

in Euclidean spacetime. He defined his metric as follows:

$$c^2 dt^2 = c^2 d\tau^2 + dx^2 + dy^2 + dz^2. \quad (6.1)$$

He also noted that the above is mathematically equivalent to

$$c^2 d\tau^2 = c^2 dt^2 - dx^2 - dy^2 - dz^2.$$

The second to discover equation (6.1) was in a study by Gill and Lindesay in 1993 (see [31]). The third to discover equation (6.1) was in the works of Bogolubov (Jr.) and Prykarpatsk. A number of papers have been published since 2010 (see [6], [7], [8], [9], [10]). They noted that they could develop an equivalent Maxwell theory in Euclidean space.

Remark 27. *There is no reference but equation (6.1) was also discovered by a senior physics major from Duke University prior to 1997. During the summer of that year, while visiting at Princeton University, he discussed plans, with the first author, to write his senior thesis on the implications of this equation.*

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- (1) They have no relevant financial or non-financial interests to disclose.
- (2) They have no conflicts of interest to declare that are relevant to the content of this manuscript.
- (3) They have no financial or proprietary interests in any material discussed in this manuscript.
- (4) They have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

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